Contents

Exercises are from Stephen Abbott's *Understanding Analysis*, Second Edition.

Problem 0: Combinatorial proof that $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$

Theorem 1: Let S be a set with n elements where $n >= 1$. Then the number of even-sized subsets of S is the same as the number of odd-sized subsets of S.

Proof: I previously proposed "flipping all bits", i.e., mapping every subset to its complement. This works for odd n: it bijectively maps odd-sized subsets to even-sized subsets and vice-versa, hence proving that the number of each is equal. But for even n the map doesn't change the parity.

Instead, let's pick some element, say $a \in S$. This is possible because $n \geq 1$. Let $T \subseteq S$ be any subset. Then define the map $f : 2^S \to 2^S$ as follows:

$$f(T) = \begin{cases} T \cup \{a\} \text{ if } a \notin T \\ T \setminus \{a\} \text{ if } a \in T. \end{cases}$$

In other words, we toggle the presence of this element a, or "flip the first bit".

This is a bijection because it is its own inverse: applying this map twice gives the original subset back.

It always changes the parity of a subset because it always changes the number of elements by 1: for subsets that contain a it removes one, and for subsets that don't it adds one.

Thus we have exhibited a bijection between odd-sized and even-sized subsets. So there are an equal number of each. ■

Problem 1: Cauchy's Remainder Theorem

Exercise 6.6.9

Let f be differentiable $N+1$ times on $(-R, R)$. For each $a \in (-R, R)$, let $S_N(x, a)$ be the partial sum of the Taylor series for f centered at a; in other words, define

$$S_N(x,a) = \sum_{n=0}^{N} c_n (x-a)^n$$

where

$$c_n = \frac{f^n(a)}{n!}.$$

Let

$$E_N(x,a) = f(x) - S_N(x,a).$$

Fix $x \neq 0$ in $(-R, R)$.

(a)

We first note that when $a = x$ the factor

$$c_0 = \frac{f(x)}{0!} = f(x).$$

Now substituting $a = x$ in the expression for E_N,

$$E_N(x,x) = f(x) - S_N(x,x) = f(x) - \sum_{n=0}^{N} c_n (x-x)^n = f(x) - c_0 = 0,$$

where all terms in the sum $S_N(x,x)$ are zero except $n = 0$.

(b)

$S_N(x,a)$ as a function of a is a finite sum of terms, each of which is the product of a polynomial $(x-a)^n$ and $f^n \frac{a}{n!}$ where $n \leq N$. By hypothesis, f is differentiable $N+1$ times, so each factor in the product is differentiable at least once. By the product and sum rules for differentiation, the entire finite sum is differentiable.

$E_N(x,a) = f(x) - S_N(x,a)$, but $f(x)$ is a constant with respect to a. So $E_N(x,a)$ is differentiable with respect to a simply because $S_N(x,a)$ is differentiable with respect to a.

For $n \geq 1$ the term with index n in the finite sum S_N is

$$\frac{f^n(a)(x-a)^n}{n!}$$

so its derivative, using the ordinary rules of differentiation, is

$$\frac{f^{n+1}(a)(x-a)^n}{n!} - \frac{f^n(a)(x-a)^{n-1}}{(n-1)!}.$$

For $n = 0$ the term is simply $f(a)$ and its derivative is $f'(a)$.

Summing over all n we find that consecutive positive and negative terms cancel, leaving

$$S'_N(x,a) = \frac{f^{N+1}(a)}{N!}(x-a)^N.$$

So

$$E'_N(x,a) = \frac{-f^{N+1}(a)}{N!}(x-a)^N.$$

(c)

Let $g(a) = E_N(x,a)$. Then g is differentiable on $(-R,R)$ and therefore also continuous on that interval. So it satisfies the hypotheses for the Mean Value Theorem for any closed interval contained in $(-R,R)$.

For $x > 0$ we apply the Mean Value Theorem to $g(a)$ on the interval $[0,x]$:

$$\frac{g(x)-g(0)}{x-0} = g'(c)$$

for some $c \in (0,x)$.

This evaluates to

$$g(x)-g(0) = xg'(c) = -x\frac{f^{N+1}(c)}{N!}(x-c)^N.$$

But $g(x) = 0$ and $g(0) = E_N(x,0)$ so

$$E_N(x,0) = x\frac{f^{N+1}(c)}{N!}(x-c)^N$$

as desired.

When $x < 0$ an exactly analogous application of MVT to the interval $[x,0]$ produces the same result.

Problem 2: Convergence of Taylor series for $1/\sqrt{1-x}$

Exercise 6.6.10

I suspect I have greatly overcomplicated this one, but I think it works. I found some identities on Wikipedia for the double factorial that I used here, and the log identity comes from the Taylor series for log from calculus that we haven't shown in this book. There must be a simpler way…

I'll use the double factorial notation

$$n!! = n \cdot (n-2) \cdot (n-4) \cdots 1$$

for odd n and

$$n!! = n \cdot (n-2) \cdot (n-4) \cdots 2$$

for even n.

Then the n'th derivative of $f(x) = 1/\sqrt{1-x}$ is

$$f^n(x) = \frac{(2n-1)!!}{2^n}(1-x)^{-\frac{2n+1}{2}}.$$

Evaluating at $x = 0$ gives the Taylor series

$$1 + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{2^n n!} x^n$$

(a)

The function is infinitely differentiable on $(-1, 1)$ so Lagrange's Remainder Theorem applies. The error term from Lagrange's Remainder Theorem is

$$E_N(x) = \frac{f^{N+1}(c)}{(N+1)!} x^{N+1}$$

for some $|c| < |x|$. This evaluates to

$$E_N(x) = \frac{(2N+1)!!}{2^{N+1}(N+1)!}(1-c)^{-\frac{2N+3}{2}} x^{N+1}.$$

For $x \in \left[0, \frac{1}{2}\right]$,

$$|x| \leq \left|\frac{1}{2}\right|$$

so

$$\left|x^{N+1}\right| \leq \frac{1}{2^{N+1}}.$$

Also, $|c| < |x|$ by the Theorem so $0 \leq c < x \leq \frac{1}{2}$ so

$$1 - c > \frac{1}{2}$$

so

$$(1-c)^{-\frac{2N+3}{2}} < 2^{\frac{2N+3}{2}}.$$

Therefore

$$|E_N(x)| < \frac{(2N+1)!!}{2^{N+1}(N+1)!} 2^{\frac{2N+3}{2}} \frac{1}{2^{N+1}} = \frac{(2N+1)!!}{(N+1)! 2^{\frac{2N+1}{2}}}.$$

Write the power-of-two factor in the denominator as

$$2^{\frac{2N+1}{2}} = \frac{2^{N+1}}{\sqrt{2}}$$

so we can multiply 2^{N+1} with the factorial.

Writing out the factorial and multiplying every term by 2 gives

$$2^{N+1}(N+1)! = 2^{N+1}(N+1) \cdot N \cdot (N-1) \cdots 1 = (2N+2) \cdot 2N \cdot (2N-2) \cdots 2 = (2N+2)!!$$

so the error term bound becomes

$$|E_N(x)| < \sqrt{2} \frac{(2N+1)!!}{(2N+2)!!}.$$

Writing out the double factorial ratio explicitly,

$$\frac{(2N+1)!!}{(2N+2)!!} = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2N+1}{2N+2} = \prod_{k=1}^{N+1} \frac{2k-1}{2k} = \prod_{k=1}^{N+1} \left(1 - \frac{1}{2k}\right).$$

Taking logarithms and using the fact that $\log(1-t) \leq -t$ for $t \in [0, 1)$,

$$\log|E_N(x)| \leq \frac{\log 2}{2} - \sum_{k=1}^{N+1} \frac{1}{2k}.$$

The sum on the RHS is $-\frac{1}{2}$ times the harmonic sum and therefore diverges to $-\infty$, so $E_N(x) \to 0$.

For $x > \frac{1}{2}$, the bound we can get from the Lagrange Remainder blows up: in the worst case $1 - c \to 1 - x$ which gives

$$E_N(x) \sim \text{all the double factorial stuff} \cdot \left(\frac{x}{1-x}\right)^{N+1} \cdot \frac{1}{\sqrt{1-x}}$$

which blows up geometrically for $x > \frac{1}{2}$ while the double factorial stuff only decays as $\sqrt{N}$.

(b)

Cauchy's Remainder Theorem gives

$$E_N(x) = \frac{f^{N+1}(c)}{N!}(x-c)^N x$$

for some c between 0 and x. This evaluates to

$$\frac{(2N+1)!!}{2^{N+1} N!}(1-c)^{-\frac{2N+3}{2}}(x-c)^N x = \frac{(2N+1)!!}{2^{N+1} N!} \cdot \left(\frac{x-c}{1-c}\right)^N \cdot (1-c)^{-\frac{3}{2}} \cdot x.$$

For $x = 0$ convergence is trivial. So we assume $x \neq 0$.

For $x \in (0, 1)$ we have

$$0 < x < 1$$

and

$$0 < c < x.$$

Write

$$\frac{x-c}{1-c}$$

as

$$\frac{1-c-(1-x)}{1-c} = 1 - \frac{1-x}{1-c}.$$

As c grows, $1-c$ becomes smaller, so $\frac{1}{1-c}$ gets bigger, so $-\frac{1}{1-c}$ gets smaller. So

$$\frac{x-c}{1-c}$$

has its maximum at $c = 0$, i.e.,

$$\frac{x-c}{1-c} \leq x$$

so

$$\left(\frac{x-c}{1-c}\right)^N \leq x^N.$$

Now,

$$0 < c < x$$

so

$$1 > 1-c > 1-x$$

so

$$1 < (1-c)^{-\frac{3}{2}} < (1-x)^{-\frac{3}{2}}.$$

Putting these bounds together we have

$$|E_N(x)| \leq \frac{(2N+1)!!}{2^{N+1}N!} \cdot x^N \cdot (1-x)^{-\frac{3}{2}} \cdot x$$

Similar to the previous part, we write

$$2^{N+1}N! = \frac{(2N+2)!!}{N+1}$$

so

$$\frac{(2N+1)!!}{2^{N+1}N!} = \frac{(2N+1)!!(N+1)}{(2N+2)!!} \leq N+1,$$

so the whole error is bounded by

$$|E_N(x)| \leq (N+1)x^{N+1}(1-x)^{-\frac{3}{2}}.$$

This goes to 0 by the Ratio Test for sequences because the ratio of consecutive terms is

$$x\frac{N+2}{N+1} \to x < 1.$$

Phew!

Problem 3: $U(f) \geq L(f)$

Exercise 7.2.1

Lemma 1: Let f be a bounded function on $[a, b]$ and let P be an arbitrary partition of $[a, b]$. Then $U(f) \geq L(f, P)$.

Proof: Let $\mathcal{P}$ be the set of all possible partitions of the interval $[a, b]$. Then by Lemma 7.2.4, for any partition $Q \in \mathcal{P}$,

$$L(f, P) \leq U(f, Q).$$

In other words, $L(f, P)$ is a lower bound for the set of all upper sums $\{U(f, Q) : Q \in \mathcal{P}\}$. But $U(f)$ is defined to be the *greatest* lower bound of the set of all upper sums, so $U(f) \geq L(f, P)$. ■

Lemma 2: For any bounded function f on $[a, b]$, it is always the case that $U(f) \geq L(f)$.

Proof: Since the partition P in the previous lemma was arbitrary, $U(f)$ is an upper bound of the set of all lower sums $\{L(f, P) : P \in \mathcal{P}\}$. But $L(f)$ is defined as the *least* upper bound of this set, so $L(f) \leq U(f)$. ■

Problem 4: Sequential Criterion for Integrability

Exercise 7.2.3

(a)

Theorem 2: A bounded function f is integrable on $[a, b]$ if and only if there exists a sequence of partitions $(P_n)_{n=1}^{\infty}$ satisfying

$$\lim_{n\to\infty} [U(f, P_n) - L(f, P_n)] = 0,$$

and in this case

$$\int_a^b f = \lim_{n\to\infty} U(f, P_n) = \lim_{n\to\infty} L(f, P_n).$$

Proof:

($\Rightarrow$)

Suppose f is integrable on $[a, b]$. Then by putting $\varepsilon = 1/n$ for every n in Theorem 7.2.8 (Integrability Criterion), we obtain a sequence of partitions P_n of $[a, b]$ such that

$$U(f, P_n) - L(f, P_n) < 1/n.$$

Also

$$U(f, P_n) \geq L(f, P_n)$$

since on every subinterval of the partition, the supremum of the function is greater than or equal to the infimum.

So

$$0 \leq U(f, P_n) - L(f, P_n) < 1/n$$

so by the Squeeze Theorem,

$$\lim_{n\to\infty} [U(f, P_n) - L(f, P_n)] = 0.$$

($\Leftarrow$)

Suppose that there is a sequence of partitions satisfying

$$\lim_{n\to\infty} [U(f, P_n) - L(f, P_n)] = 0.$$

Let $\varepsilon > 0$ be given. Then by the definition of limit, there exists some $N \in \mathbb{N}$ such that

$$|U(f, P_n) - L(f, P_n)| < \varepsilon$$

whenever $n \geq N$. Therefore

$$U(f, P_N) - L(f, P_N) < \varepsilon$$

so by the Integrability Criterion in the other direction, since ε was arbitrary, f is integrable on $[a, b]$.

Now since the function is integrable, $U(f) = L(f)$. Since $L(f)$ is the supremum of all lower sums and $U(f)$ the infimum of all upper sums,

$$L(f, P_n) \leq L(f) = U(f) \leq U(f, P_n)$$

so

$$0 \leq I - L(f, P_n) \leq U(f, P_n) - L(f, P_n)$$

and

$$0 \leq U(f, P_n) - I \leq U(f, P_n) - L(f, P_n)$$

where $I = U(f) = L(f)$ is the integral.

By the Squeeze Theorem applied to both these inequalities,

$$\lim_{n\to\infty} U(f, P_n) = \lim_{n\to\infty} L(f, P_n) = I.$$

■

(b)

For each n let P_n be the partition of $[0, 1]$ into n equal subintervals and $f(x) = x$. Then the supremum of f on the k'th subinterval is k/n and the infimum is $(k-1)/n$ for $k = 1..n$. And each subinterval has length $1/n$. So

$$U(f, P_n) = \frac{1}{n}\left[\frac{1}{n} + \frac{2}{n} + \ldots + \frac{n-1}{n} + \frac{n}{n}\right] = \frac{1}{n} \cdot \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n}$$

and

$$L(f, P_n) = \frac{1}{n}\left[\frac{0}{n} + \frac{1}{n} + \ldots + \frac{n-1}{n}\right] = \frac{1}{n} \cdot \frac{1}{n} \cdot \frac{n(n-1)}{2} = \frac{n-1}{2n}.$$

(c)

$$U(f, P_n) - L(f, P_n) = \frac{1}{n}$$

whose limit is 0 as $n \to \infty$, so by the criterion in part (a), $f(x) = x$ is integrable on $[0, 1]$. Taking the limit of $U(f, P_n)$,

$$\int_0^1 f = \lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2}.$$

Problem 5: The uniform limit of integrable functions is integrable

Exercise 7.2.5

The main idea is to bound the difference $M_k^f - m_k^f$ in terms of $M_k^{f_n} - m_k^{f_n}$, then use the Integrability Criterion in both directions.

Theorem 3: Suppose that for each n, f_n is an integrable function on $[a, b]$. If $(f_n) \to f$ uniformly on $[a, b]$ then f is also integrable on this set.

Proof: If $a = b$ then the conclusion is trivial, so we assume $b > a$.

Let $\varepsilon > 0$ be given and let

$$\delta = \frac{\varepsilon}{4(b-a)}.$$

By uniform convergence of (f_n) to f, there exists some $N \in \mathbb{N}$ such that whenever $m \geq N$, for any point $x \in [a, b]$ we have

$$|f(x) - f_m(x)| < \delta.$$

This is equivalent to the following inequalities which we will use below:

$$f(x) < f_m(x) + \delta$$

and

$$f_m(x) < f(x) + \delta.$$

Fix any $m \geq N$.

By integrability of f_m, there exists some partition P of $[a, b]$ such that

$$U(f_m, P) - L(f_m, P) < \frac{\varepsilon}{2}.$$

Let $[x_{k-1}, x_k]$ be the k'th subinterval of P and let

$$m_k^g = \inf\{g(x) : x \in [x_{k-1}, x_k]\}$$

and

$$M_k^g = \sup\{g(x) : x \in [x_{k-1}, x_k]\}$$

where g could be f or one of the functions f_n.

Then for any $x \in [x_{k-1}, x_k]$,

$$f(x) < f_m(x) + \delta \leq M_k^{f_m} + \delta$$

so $M_k^{f_m} + \delta$ is an upper bound for the set $\{f(x) : x \in [x_{k-1}, x_k]\}$. But M_k^f is the *least* upper bound of this set so

$$M_k^f \leq M_k^{f_m} + \delta.$$

Similarly,

$$f_m(x) < f(x) + \delta \le M_k^f + \delta$$

so $M_k^f + \delta$ is an upper bound for the set $\{f_m(x) : x \in [x_{k-1}, x_k]\}$, so

$$M_k^{f_m} \le M_k^f + \delta.$$

Together, these inequalities are equivalent to

$$\left|M_k^f - M_k^{f_m}\right| \le \delta.$$

By a completely analogous argument,

$$\left|m_k^f - m_k^{f_m}\right| \le \delta.$$

By the triangle inequality,

$$\begin{aligned}
M_k^f - m_k^f &= \left|M_k^f - m_k^f\right| \\
&= \left|M_k^f - M_k^{f_m} + M_k^{f_m} - m_k^{f_m} + m_k^{f_m} - m_k^f\right| \\
&\le \left|M_k^f - M_k^{f_m}\right| + \left|M_k^{f_m} - m_k^{f_m}\right| + \left|m_k^{f_m} - m_k^f\right| \\
&\le \left|M_k^{f_m} - m_k^{f_m}\right| + 2\delta \\
&\le M_k^{f_m} - m_k^{f_m} + 2\delta.
\end{aligned}$$

Let P have p subintervals. Summing over all k, we have

$$\begin{aligned}
U(f,P) - L(f,P) &= \sum_{k=1}^{p} \left(M_k^f - m_k^f\right) \Delta x_k \\
&\le \sum_{k=1}^{p} \left(M_k^{f_m} - m_k^{f_m} + 2\delta\right) \Delta x_k \\
&= \sum_{k=1}^{p} \left(M_k^{f_m} - m_k^{f_m}\right) \Delta x_k + \sum_{k=1}^{p} 2\delta \Delta x_k \\
&= U(f_m, P) - L(f_m, P) + 2\delta(b-a) \\
&< \frac{\varepsilon}{2} + 2\delta(b-a) \\
&= \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon.
\end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, by Theorem 7.2.8 (Integrability Criterion), f is integrable. ■

Problem 6: Increasing functions are integrable

Exercise 7.2.7

The main idea is that there is telescoping cancelation of the sum $U(f, P) - L(f, P)$ if the subintervals are equal.

Theorem 4: Let $f : [a, b] \to \mathbb{R}$ be increasing on the set $[a, b]$ (i.e., $f(x) \leq f(y)$ whenever $x < y$). Then f is integrable on $[a, b]$.

Proof: If $b = a$ then the theorem is trivial so we will assume $b > a$.

Let $\varepsilon > 0$ be given and let n be any integer satisfying

$$n > \frac{(f(b) - f(a))(b - a)}{\varepsilon}.$$

This n must be positive because $f(b) \geq f(a)$.

Consider the partition P with n equal subintervals. On each subinterval $[x_{k-1}, x_k]$, the function has its supremum at x_k and its infimum at x_{k-1} because $f(x_k) \geq f(x)$ for all $x \leq x_k$, and $f(x_{k-1}) \leq f(x)$ for all $x \geq x_{k-1}$. So the supremum of one subinterval is the infimum of the next.

Also, because the subintervals are equal, let $\Delta x = x_k - x_{k-1} = \frac{b-a}{n}$ be the common length of all subintervals.

Then

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{k=1}^{n} (M_k - m_k) \Delta x \\ &= \Delta x (M_1 - m_1 + M_2 - m_2 + ... + M_n - m_n) \\ &= \Delta x (M_n - m_1) \\ &= \Delta x (f(b) - f(a)) \\ &= \frac{b - a}{n} (f(b) - f(a)) \\ &< \varepsilon. \end{aligned}$$

Since $f(a) \leq f(x) \leq f(b)$ for $x \in [a, b]$, the function is bounded, so Theorem 7.2.8 applies. Since $\varepsilon > 0$ was arbitrary and we have exhibited a partition with $U(f, P) - L(f, P) < \varepsilon$, the function is integrable. ■

Problem 7: Riemann-integrability of Dirichlet's function

Is Dirichlet's function from Section 4.1 integrable? (It's the function g defined on p. 112.)

Theorem 5: Dirichlet's function, defined as

$$g(x) = \begin{cases} 1 \text{ if } x \in \mathbb{Q} \\ 0 \text{ if } x \notin \mathbb{Q} \end{cases}$$

is not Riemann-integrable on any interval $[a, b]$ of $\mathbb{R}$ with $b > a$.

Proof: Let P be any partition of $[a, b]$ and let $A = [x_{k-1}, x_k]$ be any subinterval of P. (The subinterval exists and $x_k > x_{k-1}$ because $b > a$). By the density of both the rationals and irrationals in $\mathbb{R}$, there exists a rational number $q \in A$ and an irrational number $p \in A$.

Let

$$m_k = \inf\{g(x) : x \in [x_{k-1}, x_k]\}$$

and

$$M_k = \sup\{g(x) : x \in [x_{k-1}, x_k]\}.$$

Then $m_k \leq 0$ since $g(p) = 0$, but because $0 \leq g(x) \leq 1$ for all x we have $m_k = 0$. Similarly, $M_k = 1$ because $g(q) = 1$. Thus the lower sum $L(g, P) = 0$ and the upper sum $U(g, P) = \sum_{k=1}^{n} (x_k - x_{k-1}) = b - a$. That is, they don't depend on the particular partition P. Hence upper integral of g,

$$U(g) = \inf\{U(g, P) : P \in \mathcal{P}\} = \inf\{b - a\} = b - a$$

and similarly $L(g) = 0$. So $U(g) \neq L(g)$, that is, g is not Riemann-integrable. ∎